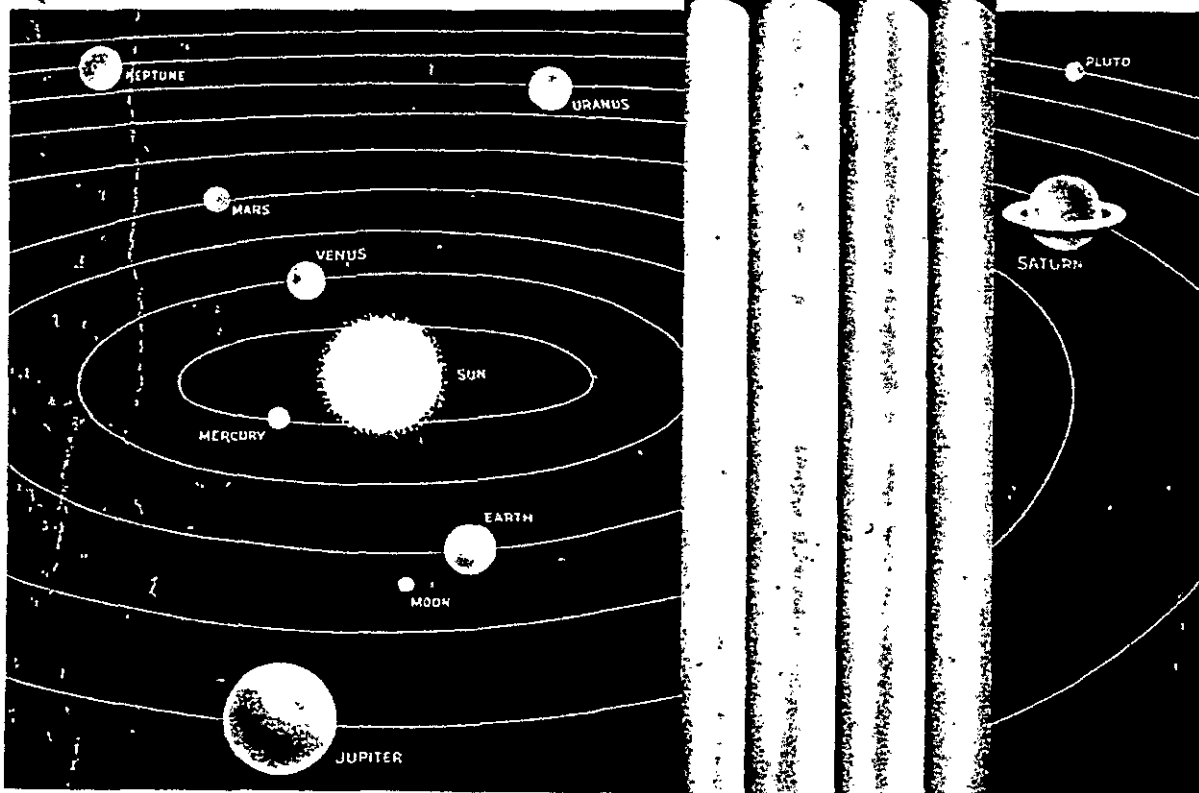


PROJECT HORIZON

Volume I SUMMARY AND SUPPORTING CONSIDERATIONS

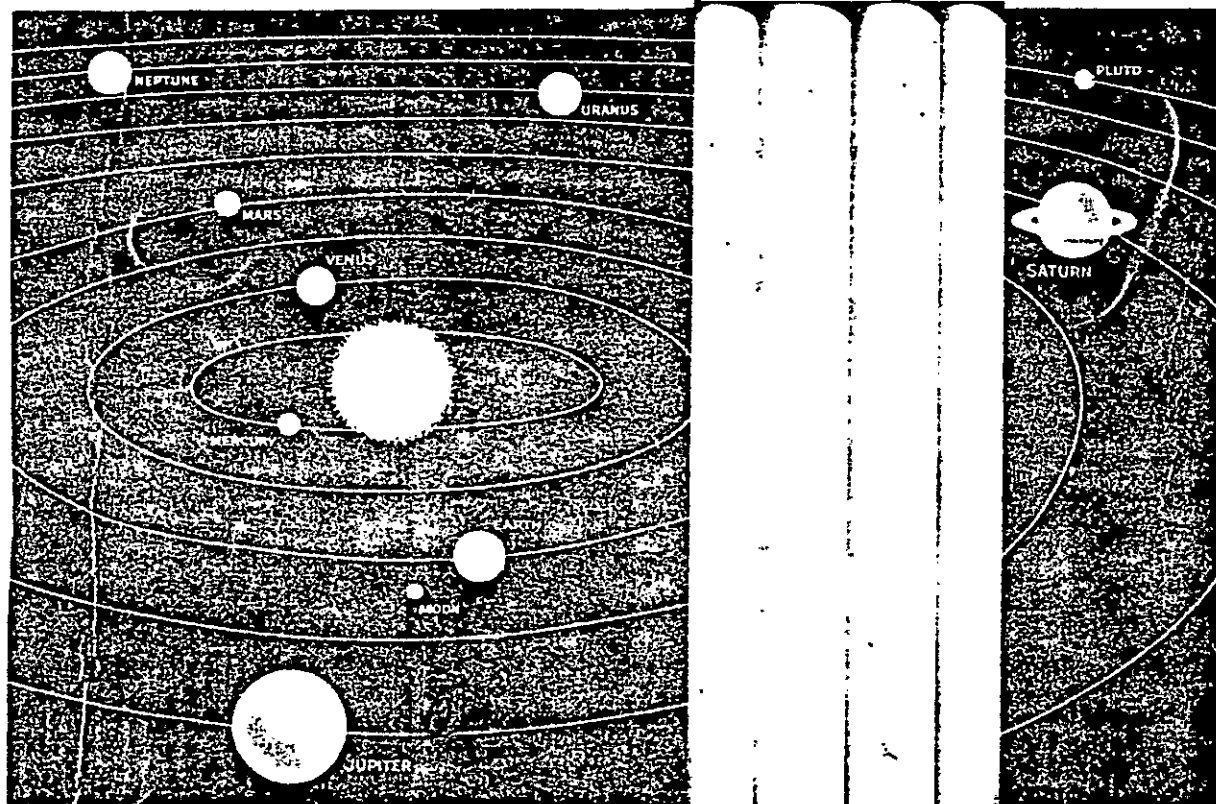


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PROJECT HORIZON

Volume II TECHNICAL CONSIDERATIONS & PLANS



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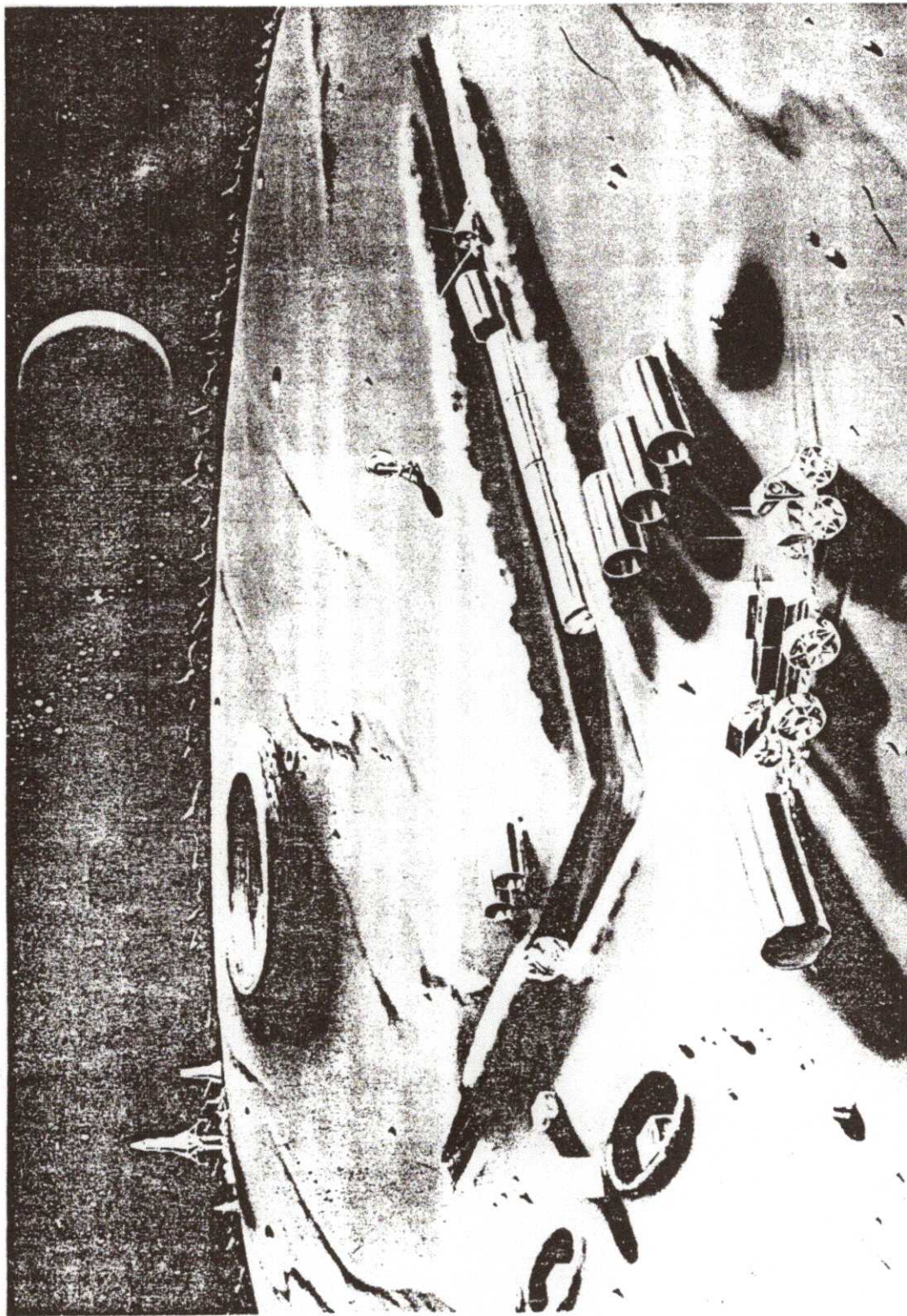


Fig. I-1. HORIZON Outpost in Late 1965

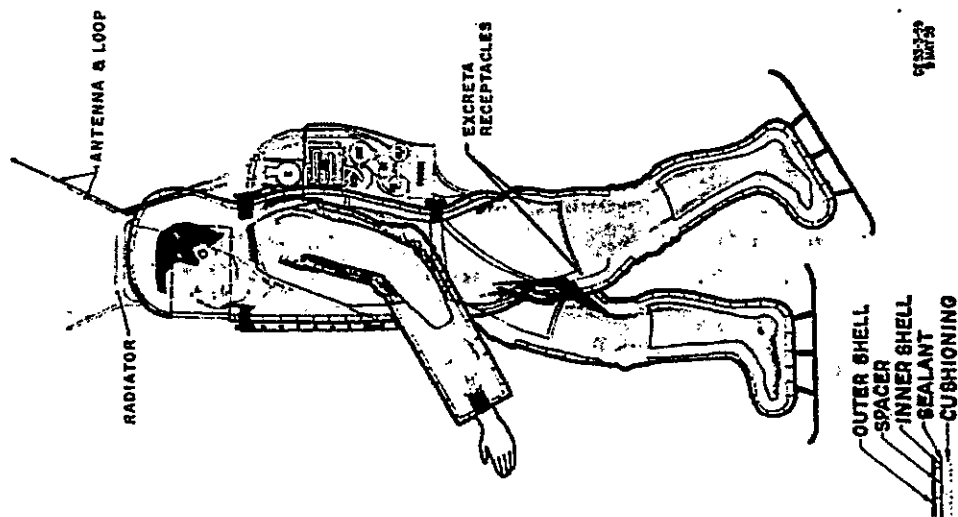
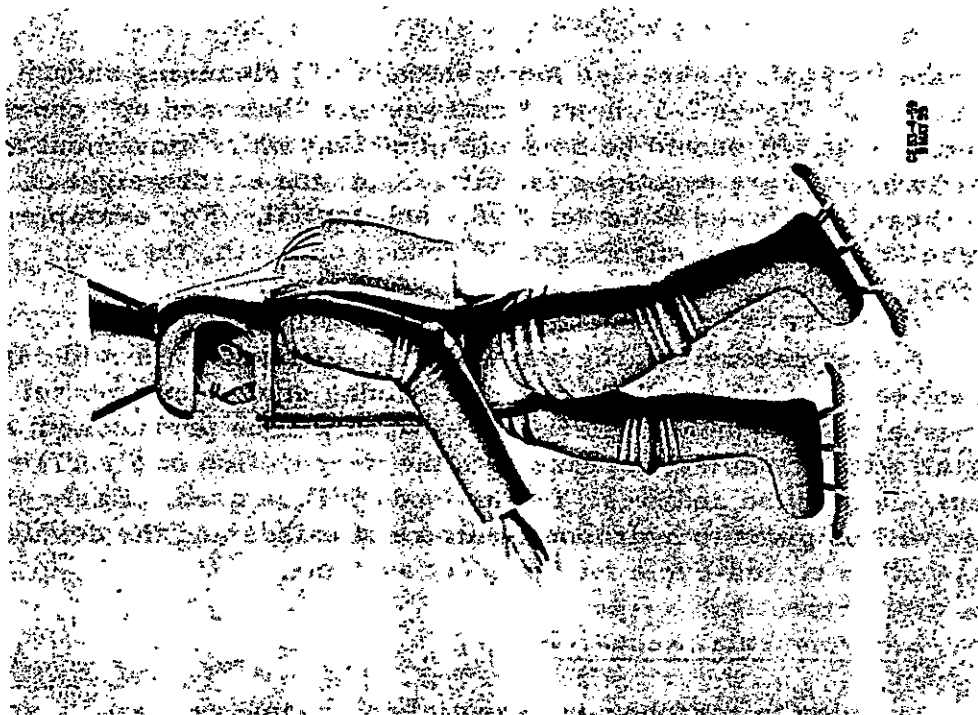


Fig. 1-5. Typical Lunar Suit

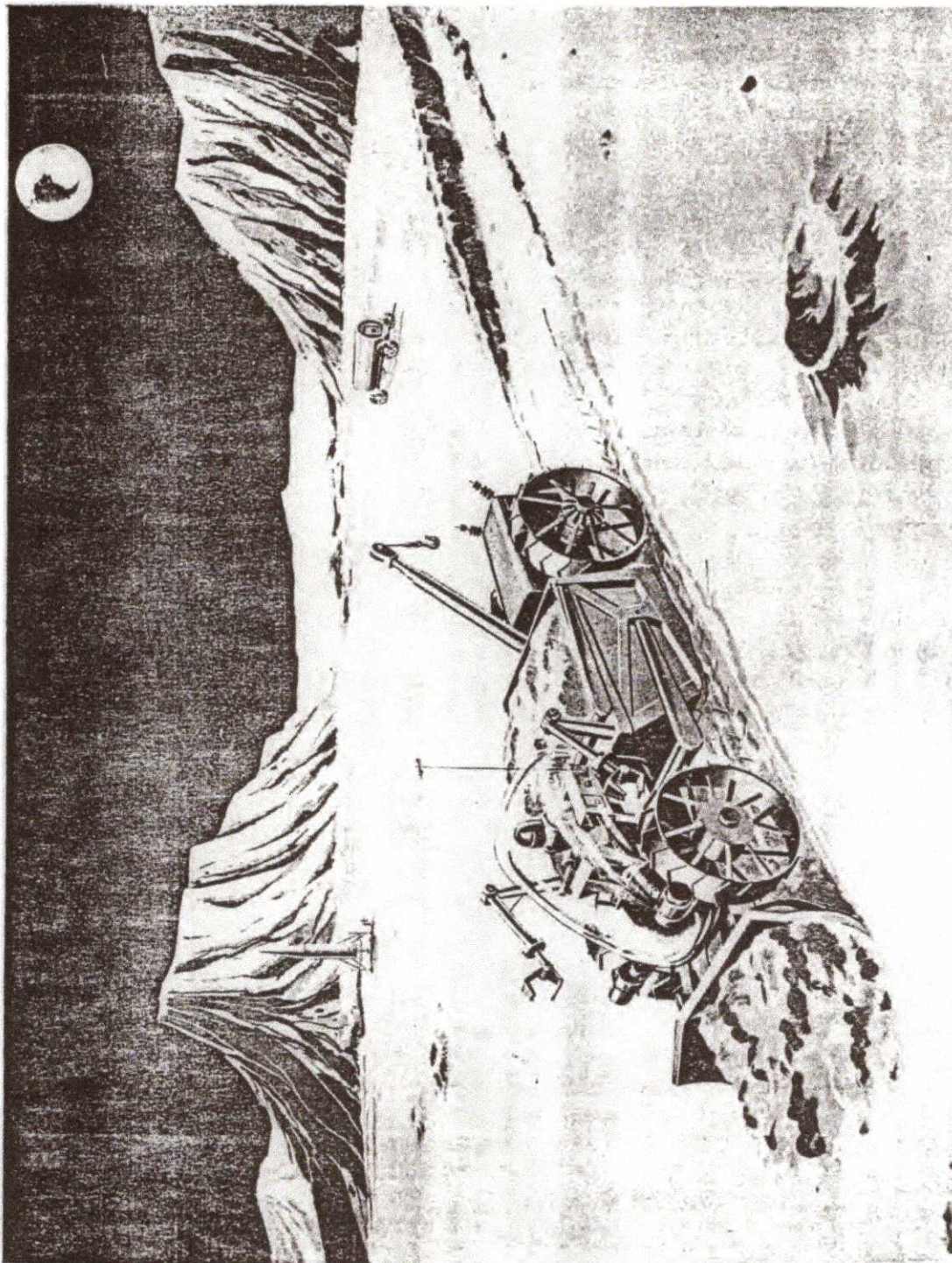


Fig. II-6. Typical Lunar Construction Vehicle

ADVANCE PARTY OUTPOST

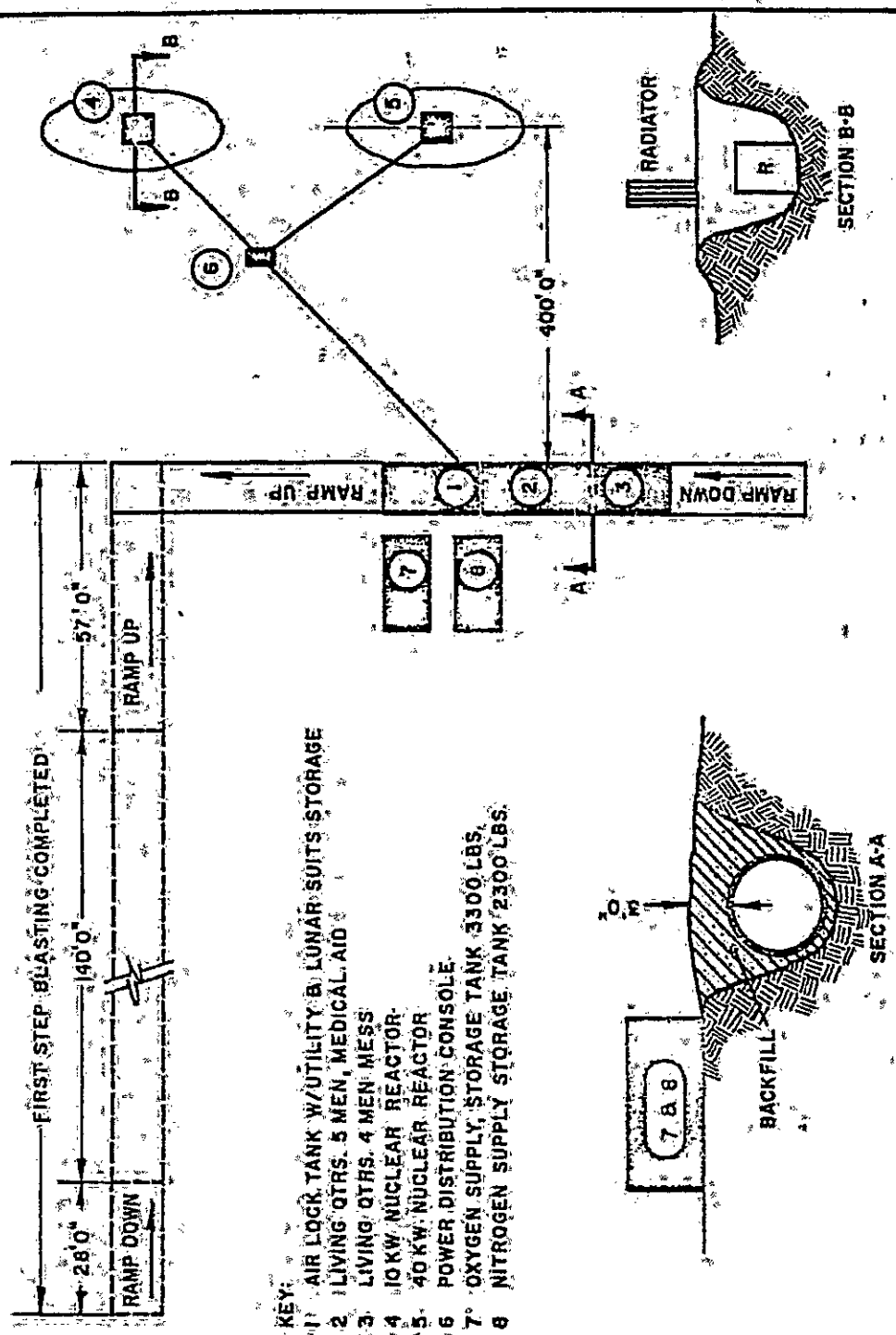


Fig. II-3. Layout of Initial Construction Camp

AIR LOCK & LIVING QUARTERS

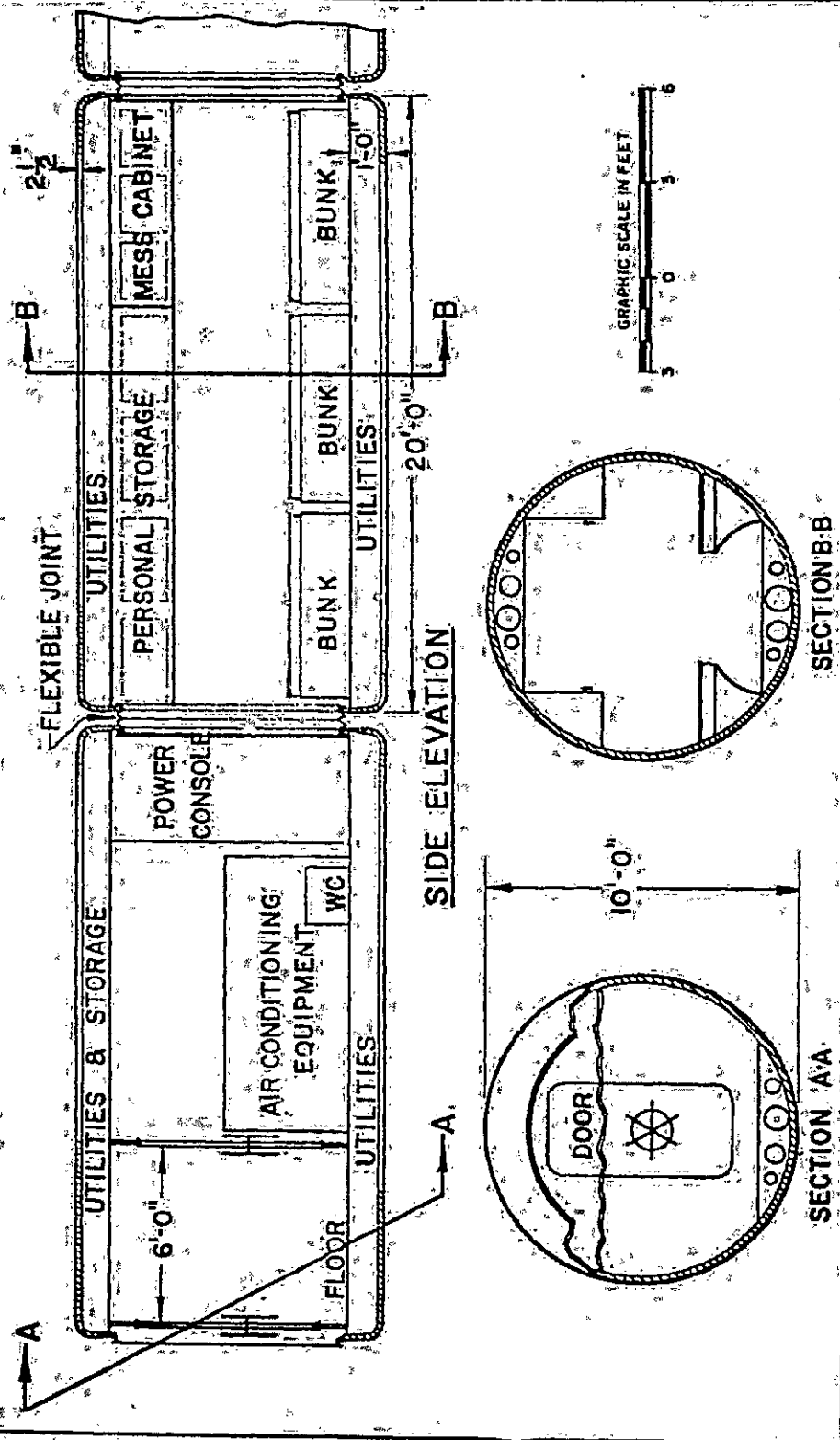


Fig. II-4. Cross-Section of Typical Outpost Compartments

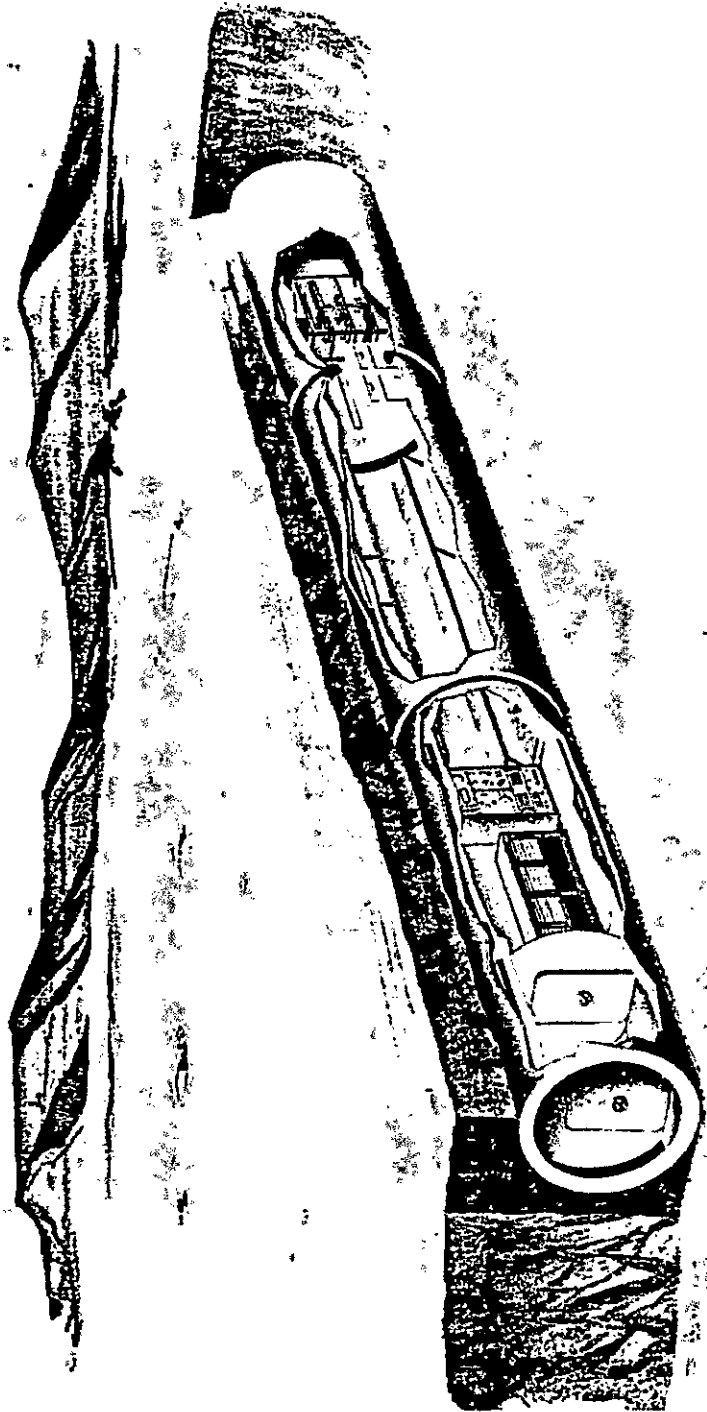
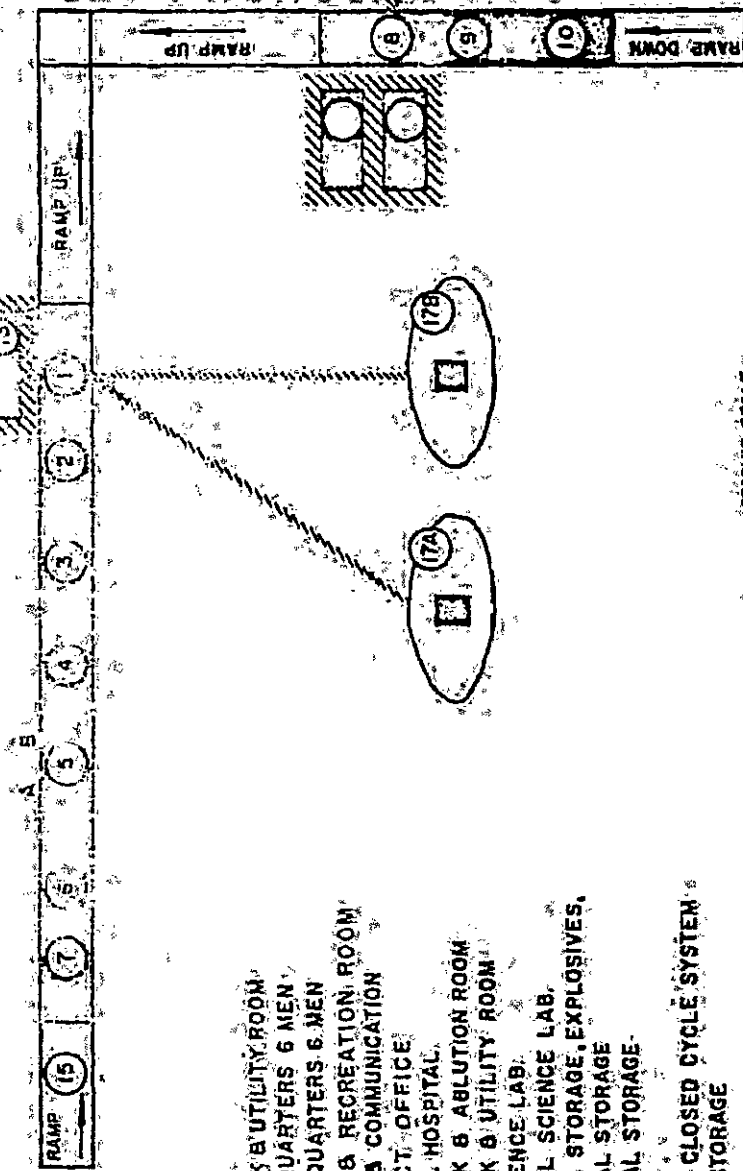


Fig. II-5. Overall View of Initial Construction Camp

BASIC OUTPOST

(12 MEN)



KEY

- 1 AIR LOCK & UTILITY ROOM
- 2 LIVING QUARTERS 6 MEN
- 3 LIVING QUARTERS 6 MEN
- 4 DINING & RECREATION ROOM
- 5A SIGNAL & COMMUNICATION
- 5B PROJECT OFFICE
- 6 MEDICAL HOSPITAL
- 7 AIR LOCK & ABILITY ROOM
- 8 AIR LOCK & UTILITY ROOM
- 9 BIO. SCIENCE LAB.
- 10 PHYSICAL SCIENCE LAB.
- 11 SPECIAL STORAGE, EXPLOSIVES
- 12 CHEMICAL STORAGE
- 13 CHEMICAL STORAGE
- 14 REEFER
- 15 FUTURE CLOSED CYCLE SYSTEM
- 16 WASTE STORAGE
- 17A 60 KW
- 17B 5 KW
- 17C 40 KW
- 17D 10 KW

REACTORS

Fig. II-7. Layout Basic 12-Man Outpost

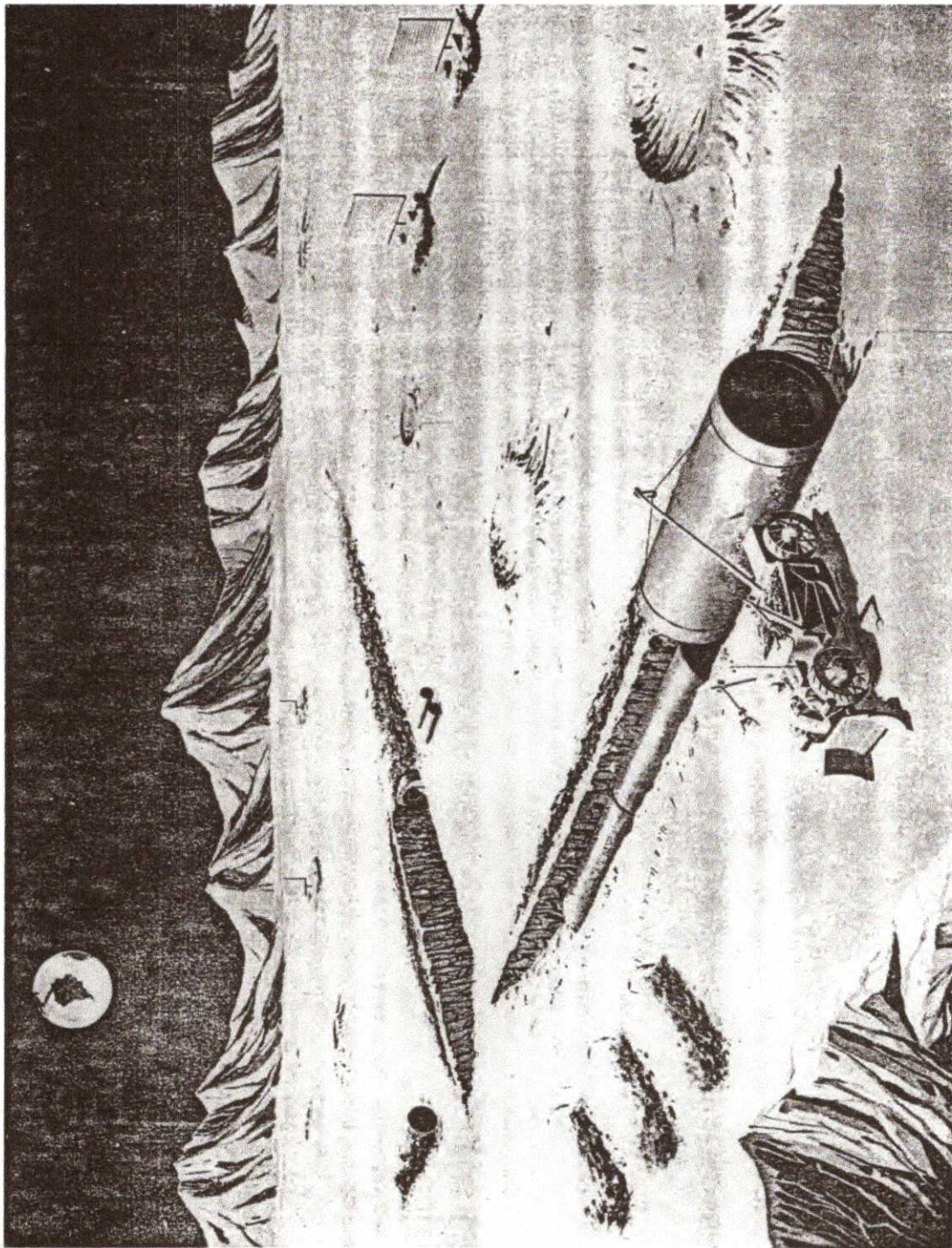


Fig. II-8. Overall View of 12-Man Outpost



Fig. II-9. Nuclear Power Plant on Moon (60 kw)

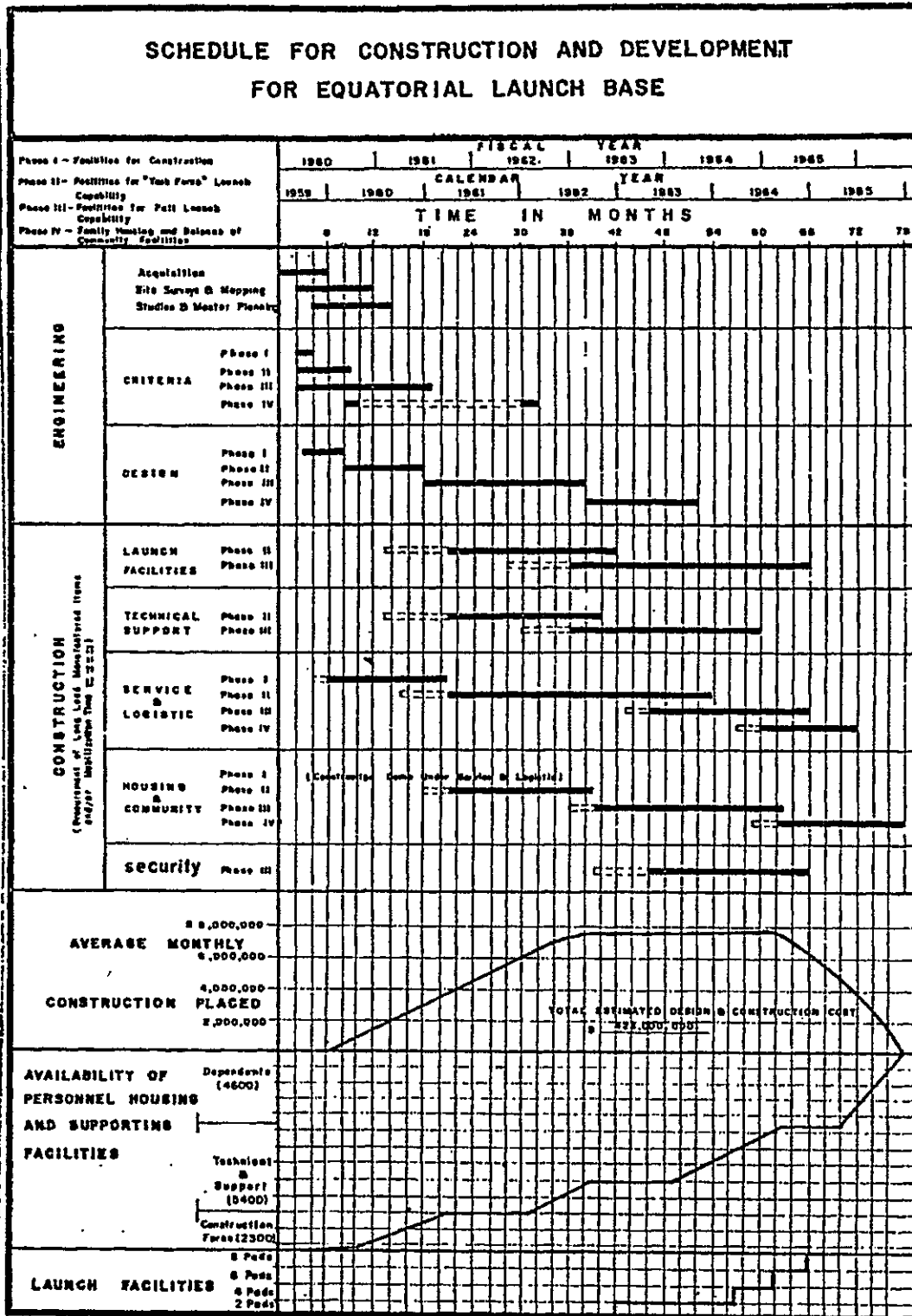


Fig. II-73. Schedule for Construction and Development for Equatorial Launch Base

TABLE II-41
SUMMARY OF HORIZON ACCOMPLISHMENTS

SUMMARY OF PLANNED ACCOMPLISHMENTS									
1959	1960	1961	1962	1963	1964	1965	1966		
1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4
Initial Funding Requirements									
First SATURN I Booster Flight									
Environment Test Facility Initial Capability									
First Lunar Soft Landing									
First Manned Lunar Circumnavigation									
First Manned Lunar Landing									
First Man Returned from Moon									
Twelve Man Lunar Outpost Completed									

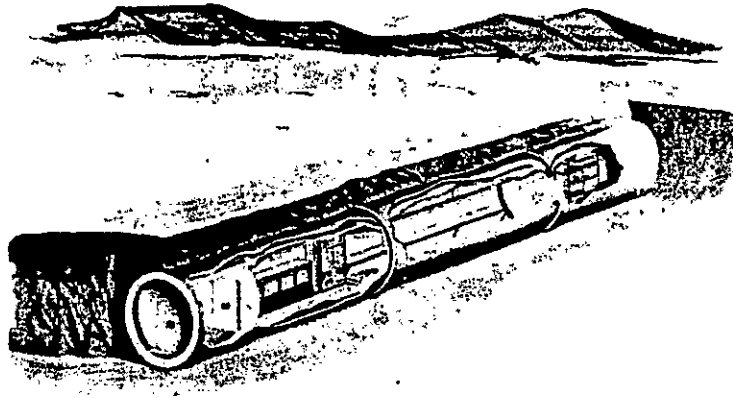


Fig. I-2. Cross Section of Typical Outpost Compartments

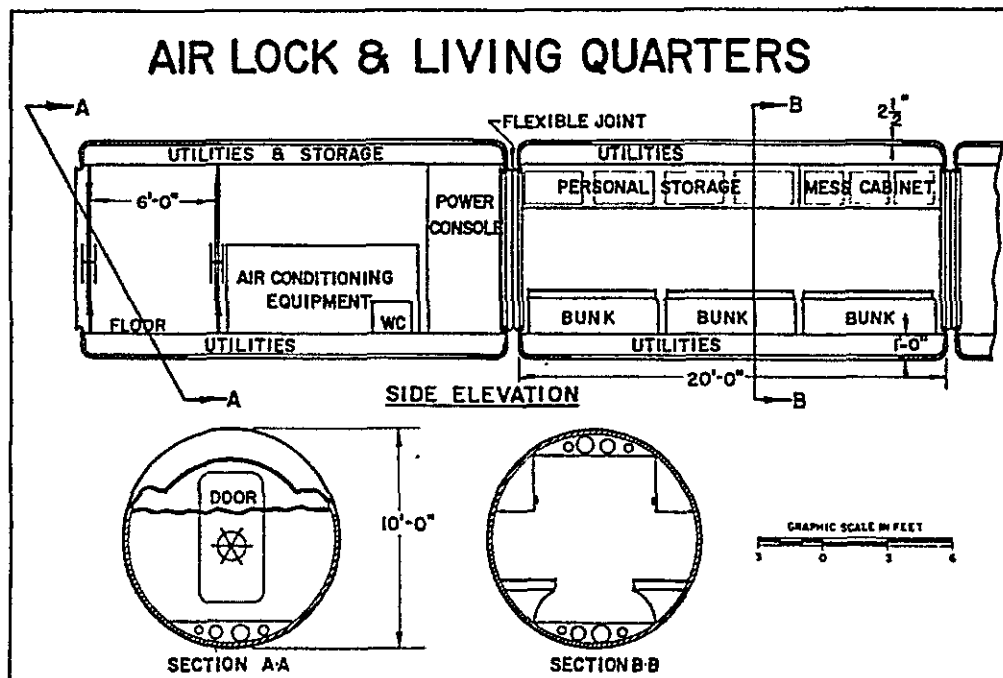


Fig. I-3. Overall View of Initial Construction Camp

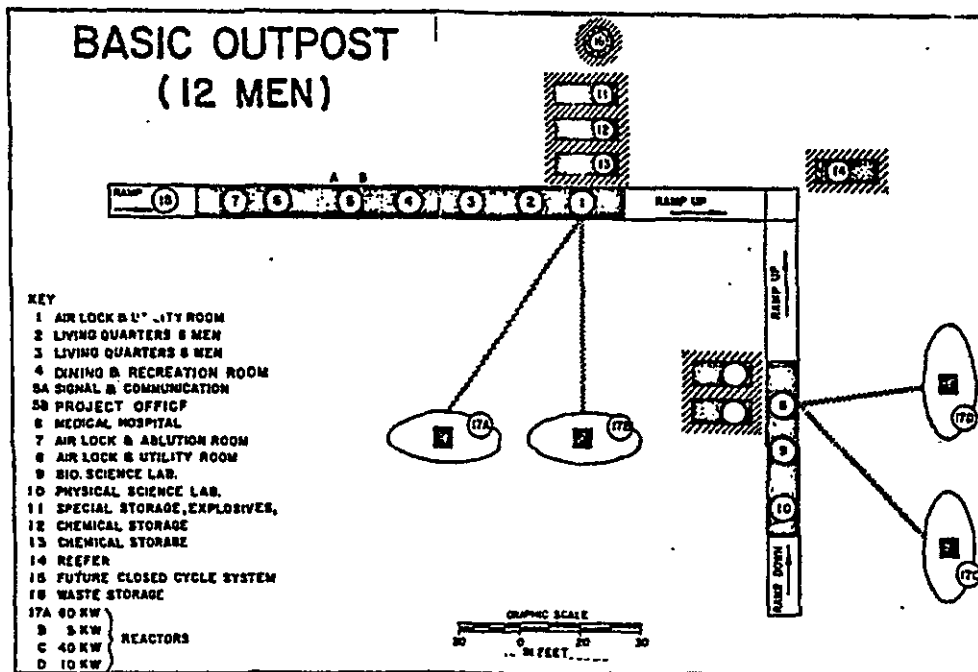


Fig. I-4. Layout of Basic 12-Man Outpost

incidental heat given off by an adequate internal lighting system will nominally supply essentially all of the heat required to maintain comfortable "room" temperature in the outpost quarters.

A suitable atmosphere will be provided within the quarters. The basic gas supply will stem from special insulated tanks containing liquid oxygen or nitrogen. The nitrogen supply needs only to provide for initial pressurization and replacement of leakage losses; whereas, the oxygen is, of course, continuously used to supply bodily needs. However, the weights and volumes of both gases are quite reasonable and presents no unusual problem of supply. Carbon dioxide and moisture will be controlled initially by a solid chemical absorbent and dehumidifier. Such a scheme requires considerable amounts of material; therefore, a carbon dioxide freeze-out system will be installed later.

4. Personnel Equipment

For sustained operation on the lunar surface a body conformation suit having a substantial outer metal surface is considered a necessity for several reasons: (1) uncertainty that fabrics and elastomers can sustain sufficient pressure differential without unacceptable leakage; (2) meteoroid protection; (3) provides a highly reflective surface;

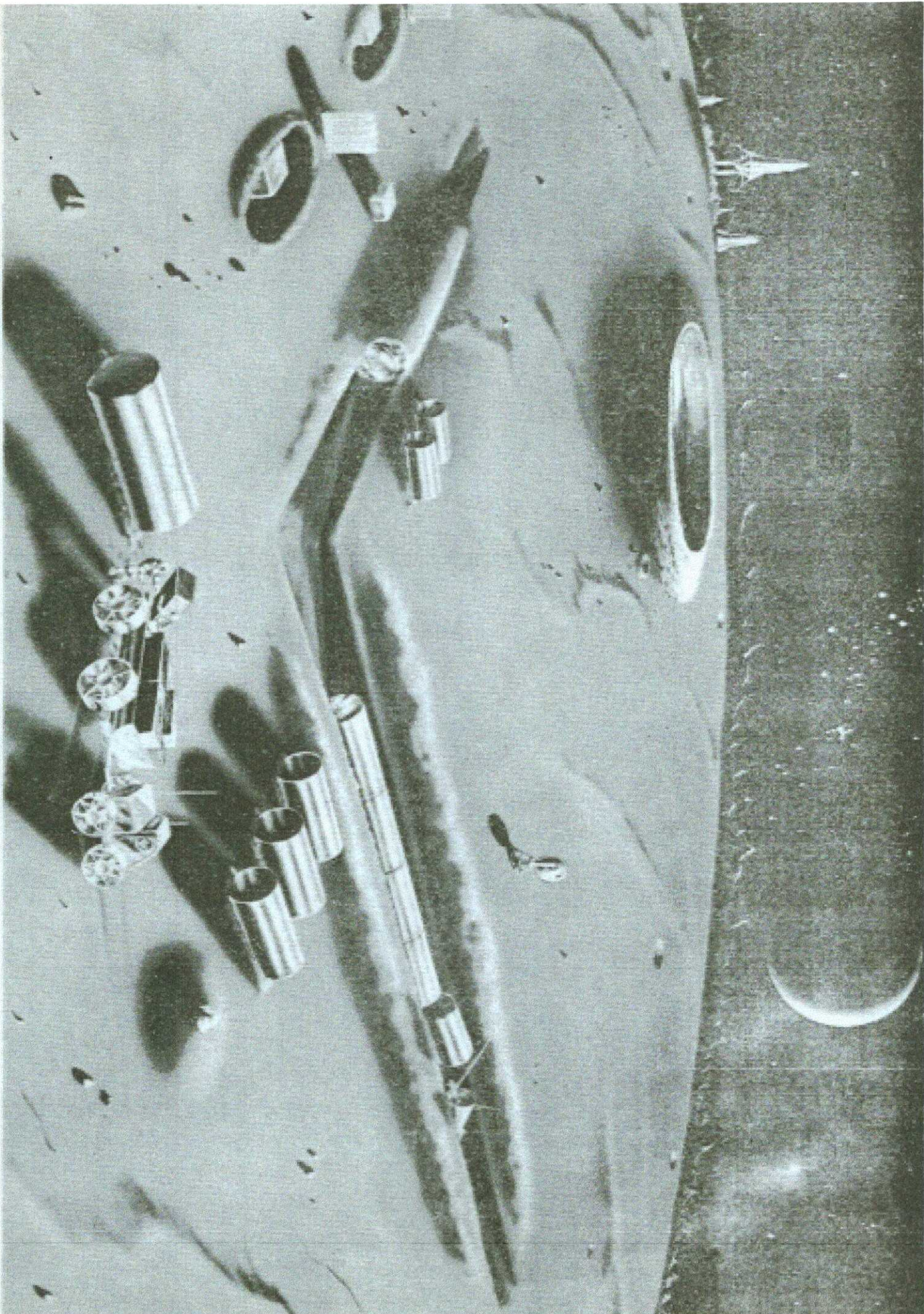


Fig. I-1. HORIZON Outpost in Late 1965